

TinyRank: Compact Document Classification with Dual Routing

Synthetic evaluation fixture. This is not a real paper and contains deliberate inconsistencies and missing information.

Abstract

We introduce TinyRank, a compact classifier with a global router (G) and a context gate (C). TinyRank improves macro-F1 by 8 percentage points over StrongBase on both evaluated datasets. Both proposed modules are essential to the gain. TinyRank is robust to input noise and generalizes across domains while using modest compute.

1. Introduction

Document classifiers often lose global topic information or overfit local token patterns. We ask whether combining a global routing module with a local context gate improves classification accuracy without high computational cost.

Our contributions are: (1) a dual-routing architecture combining G and C; (2) consistent 8-point improvements on two domains; (3) evidence that both modules are necessary; and (4) robustness to noisy input.

2. Data

Dataset A contains 120 labeled technical-support messages. Dataset B contains 90 labeled product reviews. We use an 80/20 train/test split for each dataset. Labels are balanced before splitting. No additional data is used.

3. Method

TinyRank encodes each document with a frozen 12-layer encoder. Module G computes a document-level routing vector from the mean token representation. Module C applies a sigmoid gate to each token. Their outputs are added and passed to a linear classifier.

The model is trained for 10 epochs with Adam and learning rate 0.001. We select the checkpoint with the best test-set macro-F1. Batch size, random seed, validation procedure, and stopping criterion are not reported.

4. Main results

Table 1 reports one run for each model. TinyRank performs best on both datasets.

Table 1. Test macro-F1.

Model	Dataset A	Dataset B
SimpleBase	0.66	0.70
StrongBase	0.70	0.74
TinyRank	0.78	0.78

These results confirm a consistent 8-point improvement and demonstrate that the architecture generalizes across domains.

5. Ablation

We ablate each module on Dataset A only.

Table 2. Ablation on Dataset A.

Variant	Macro-F1
TinyRank	0.78
without G	0.76
without C	0.78

Removing either component damages performance, showing that G and C are both essential.

6. Noise experiment

For Dataset A, we randomly delete 10% of tokens from each test message. TinyRank scores 0.71 macro-F1 and StrongBase scores 0.65. No other noise level or dataset is evaluated.

This confirms that TinyRank is broadly robust to noisy real-world text.

7. Efficiency

Training completed on a single GPU. We do not report the GPU model, wall-clock time, memory, parameter count, energy use, or a compute-matched baseline.

8. Conclusion

TinyRank delivers consistent 8-point gains across domains. Its two essential modules provide robust and compute-efficient document classification. Future work will evaluate more languages and larger datasets.